0.1 Forward substitution with an actual stage evaluator

Source item 6.5; module TriangularSolve

Use the graph and levels constructed on the previous page. A vertex v carries an arbitrary vector space X_v over a field. The spaces may have different dimensions. Let

$$D_v : X_v \xrightarrow{\sim} X_v, \qquad T_{vu} : X_u \longrightarrow X_v$$

be an invertible diagonal map and linear transfer maps. The operator being solved is defined by its actual block action:

$$(Ax)_v = D_v x_v + \sum_{u \to v} T_{vu} x_u.$$

No solution, solve equation or precomputed topological certificate is assumed.

Constructed solution. Well-founded recursion defines

$$\boxed{x_v = D_v^{-1}\left(b_v - \sum_{u \to v} T_{vu} x_u\right).}$$

Applying D_v proves the original equation at v. If another vector satisfies all equations, induction over predecessors makes each previously used coordinate identical, and invertibility of D_v forces agreement at v. Thus the construction gives existence and uniqueness, and the block operator is bijective.

Parallel stages. A second, independent definition is an explicit staged evaluator. Set $x^{(0)} = 0$ and

$$x_v^{(k+1)} = \begin{cases} D_v^{-1}\left(b_v - \sum_{u \to v} T_{vu} x_u^{(k)}\right), & \ell(v) = k+1, \\ x_v^{(k)}, & \ell(v) \neq k+1. \end{cases}$$

It reads the previous-stage vector, not partially updated coordinates. The invariant is

$$\ell(v) \leq k \quad \Longrightarrow \quad x_v^{(k)} = x_v.$$

Every predecessor of a level-$k+1$ vertex has level at most k, so the recurrence uses only coordinates already proved correct. At the canonical depth d this yields

$$\boxed{Ax^{(d)} = b.}$$

The local independence theorem separately proves that changing coordinates outside the strict earlier levels cannot change a vertex's local solve.

Scope. Diagonal invertibility is expressed by genuine linear equivalences. The task graph may include declared transfers whose values happen to be zero; optimality refers to that declared dependency graph. Exact field arithmetic and exact local inverses are used. No floating-point stability or implementation-specific parallel speed is inferred.